BASIC ELECTRONICS
(24B11EC111)
Unit-1
Lecture-7
(Superposition Theorem)

CONTENTS

Network Theorems

- To handle the complex circuits, engineers have developed some theorems like Superposition Theorem, Thevenin’s Theorem, Norton’s Theorem, Maximum Power Transfer Theorem, Source Transformation.
- Since these theorems are applicable to linear circuits, we first discuss the concept of circuit linearity.

Linearity Property

- Linearity is the property of an element describing a linear relationship between cause and effect.
- The property is a combination of both the **homogeneity (scaling)** property and the **additivity** property.
- The **homogeneity property** requires that if the **input is multiplied by a constant**, then the **output is multiplied** by the **same constant**.
- The **additivity property** requires that the **response to a sum of inputs** is the **sum of the responses to each input applied separately**.
- **Resistor is a linear element** because the **voltage-current relationship** satisfies both the **homogeneity** and the **additivity** properties.
- In general, a circuit is linear if it is both additive and homogeneous.
- A **linear circuit** consists of only **linear elements, linear dependent sources, and independent sources**.

Superposition Theorem

- The **superposition** principle states that **the voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltages across (or currents through) that element due to each independent source acting alone.**

- **To apply** the superposition principle, we must keep **two things** in mind:

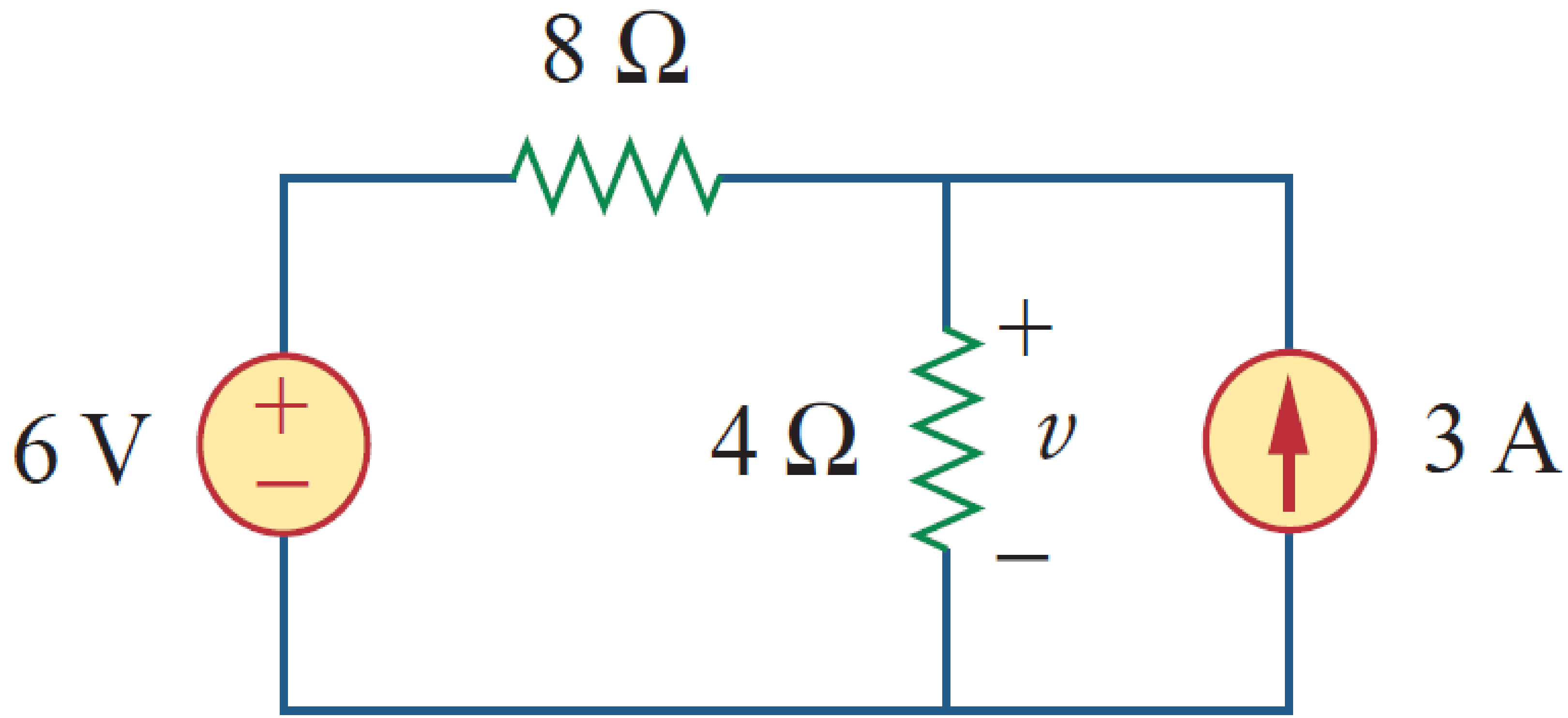
1. We consider **one independent source at a time** while all **other independent sources are turned off**. This implies that we replace **every voltage source by 0 V** (**or a short circuit**), and every **current source by 0 A (or an open circuit**). This way we obtain a simpler and more manageable circuit.
2. **Dependent sources** are **left intact** because they are **controlled by circuit variables**.

Superposition Theorem

- **With these in mind, we apply the superposition principle in three steps:**
  1. **Turn off** all **independent sources except one source.** Find the **output (voltage or current) due to that active source**.
  2. **Repeat** step 1 for each of the other independent sources.
  3. Find the **total contribution by adding algebraically all the contributions due to the independent sources**.

Problem

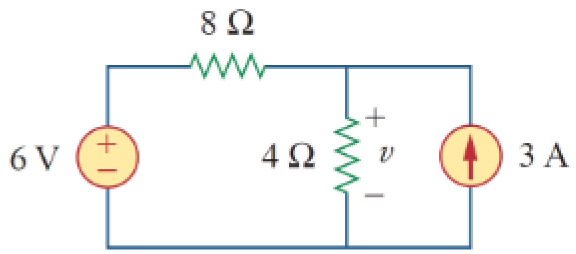
Use the superposition theorem to find v in the circuit of Fig.



Problem - Solution

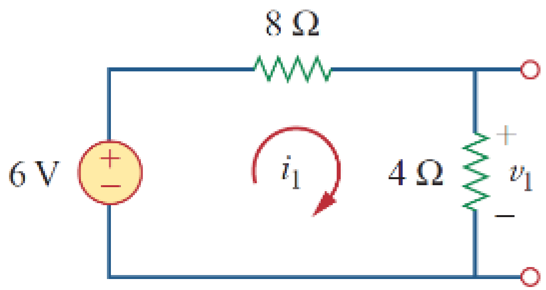
Since there are two sources, let

$$v = v_1 + v_2$$



where $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$ are the contributions due to the **6-V voltage** source and the **3-A current source**, respectively.

To obtain $\boldsymbol{v_1}$ we set the **current source to zero**, as shown in Figure.



$$12i_1 - 6 = 0 \quad \Rightarrow \quad i_1 = 0.5 \text{ A}$$

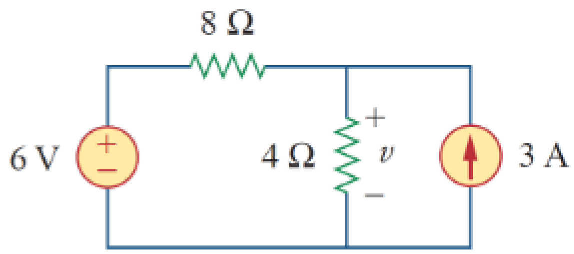
Applying KVL to the loop gives

Thus

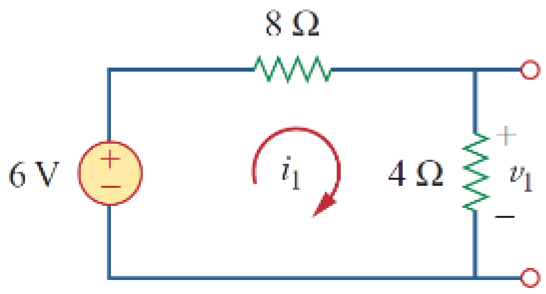
$$v_1 = 4i_1 = 2 \text{ V}$$

Problem - Solution

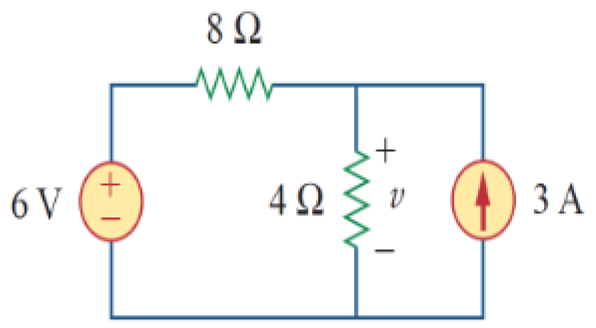
We may also use voltage division to get v_1 by writing



$$v_1 = \frac{4}{4+8}(6) = 2 \text{ V}$$



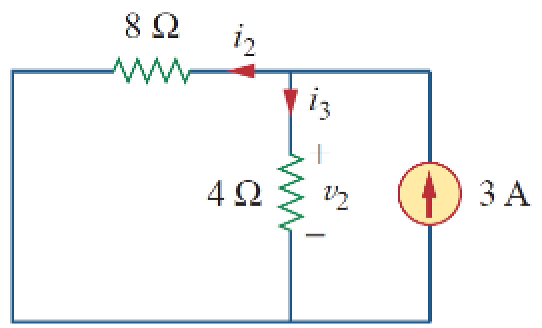
Problem - Solution



To get v_2 we set the voltage source to zero, as in Figure.

Using current division,

$$i_3 = \frac{8}{4+8}(3) = 2 \text{ A}$$



Hence,

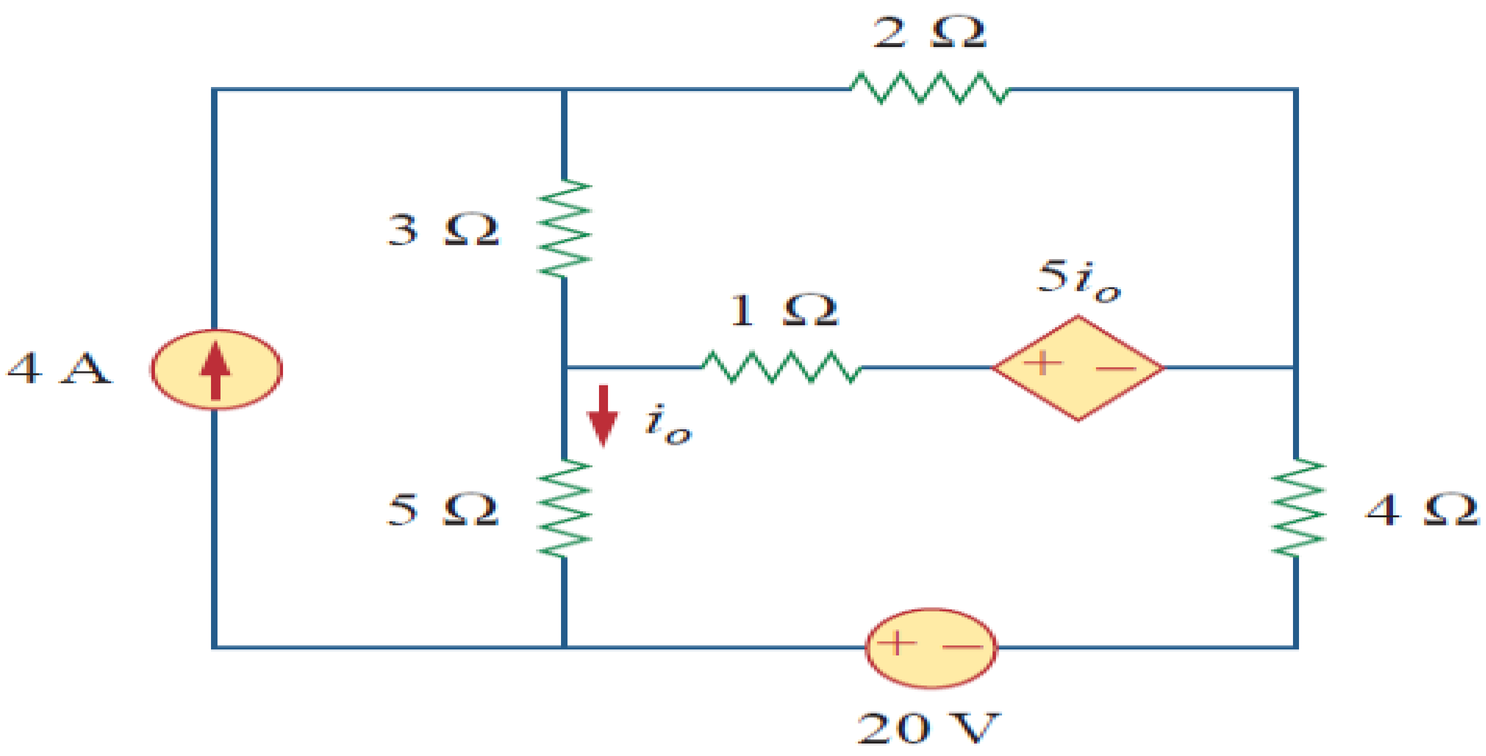
$$v_2 = 4i_3 = 8 \text{ V}$$

And we find

$$v = v_1 + v_2 = 2 + 8 = 10 \text{ V}$$

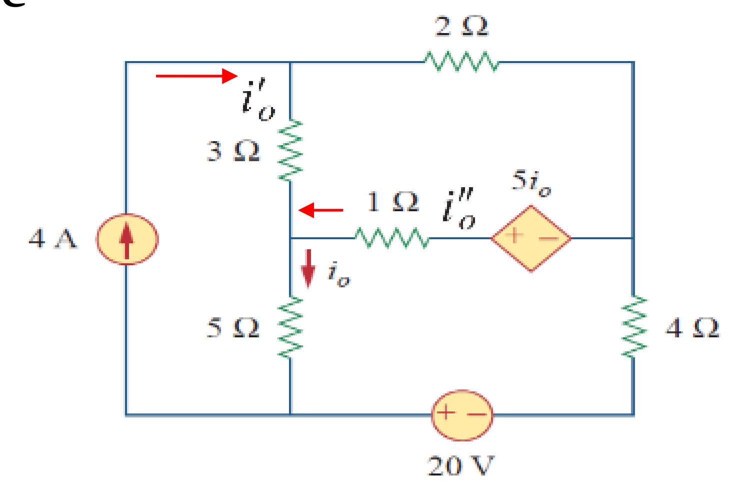
Problem

Find i_0 in the circuit of Figure using superposition.



Problem - Solution

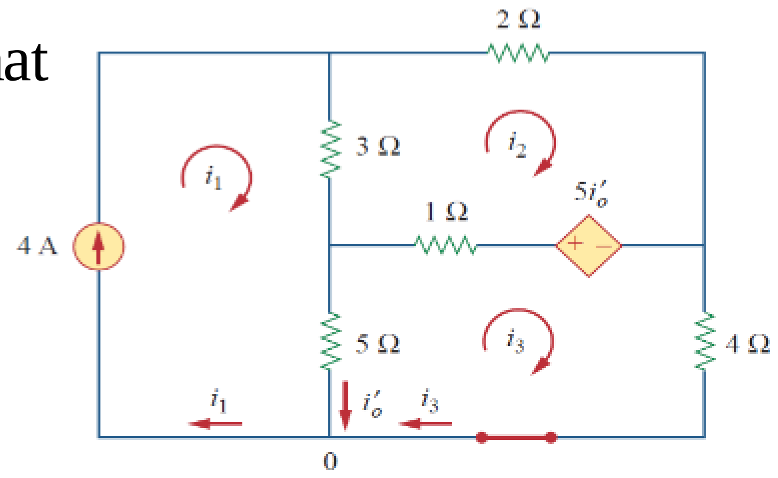
The circuit in Figure involves a **dependent source**, which must be left intact.



We let $$i_o = i'_o + i''_o$$

where i'_o and i''_o are due to the **4-A current source** and **20-V voltage source** respectively.

To obtain i'_o we **turn off the 20-V source (short circuit)** so that we have the circuit in Figure.



We apply mesh analysis in order to obtain

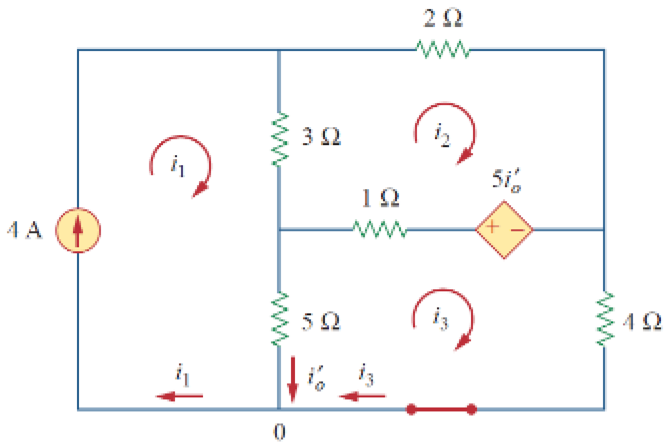
For loop 1, $$i_1 = 4 \text{ A}$$

For loop 2, $$-3i_1 + 6i_2 - 1i_3 - 5i'_o = 0$$

Problem - Solution

For loop 3, $-5i_1 - 1i_2 + 10i_3 + 5i'_o = 0$

But at node 0, $i_3 = i_1 - i'_o = 4 - i'_o$



By substituting, we get

$$3i_2 - 2i'_o = 8$$

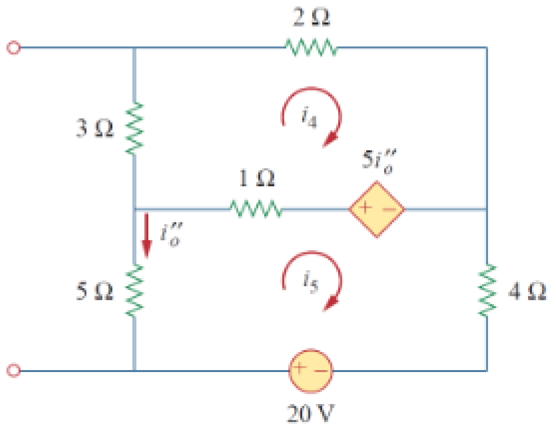
$$i_2 + 5i'_o = 20$$

After solving, we get

$$i'_o = \frac{52}{17}\ \text{A}$$

Problem - Solution

To obtain i_o'' we turn off the 4-A current source so that the circuit becomes that shown in Figure.



For loop 4, KVL gives

$$6i_4 - i_5 - 5i_o'' = 0$$

and for loop 5,

$$-i_4 + 10i_5 - 20 + 5i_o'' = 0$$

But

$$i_5 = -i_o''$$

Substituting this in Equations gives

$$6i_4 - 4i_o'' = 0$$

$$i_4 + 5i_o'' = -20$$

which we solve to get

$$i_o'' = -\frac{60}{17}\ \text{A}$$

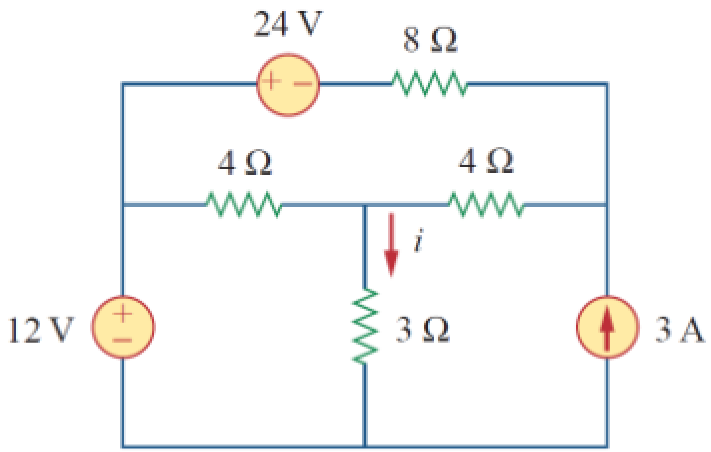
Finally, we get

$$i_o = i_o' + i_o''$$

$$i_o = -\frac{8}{17} = -0.4706\ \text{A}$$

Practice Problem

For the circuit in Figure, use the superposition theorem to find *i*.



Ans: 2 A

References

1. R.C. Dorf and James A. Svoboda, “Introduction to Electric Circuits”, 9^{th} ed, John Wiley & Sons, 2013.
2. D.C. Kulshreshtha, Basic Electrical Engineering, Revised 1^{st} ed, Tata Mc Graw Hill, 2017.
3. V. K.Mehta, Rohit Mehta, Basic Electrical Engineering, 6^{th} ed, S. Chand Publishing, 2012.